

OM RAHEJA

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Education

University of Illinois Urbana-Champaign (UIUC)

Expected May 2029

Bachelor of Science in Computer Engineering

Urbana, IL

Technical Skills

Languages: Python, JavaScript, SQL, Go, C, x86 Assembly, GDScript, HTML/CSS

Frameworks & Libraries: Next.js, React, Node.js, Express, PostgreSQL, Groq API, Gemini API, TensorFlow.js, Axios

AI & Systems: Agentic AI, Vision-Language Models (VLMs), OCR, Structured Data Extraction, Multi-Agent Systems, OpenClaw, AutoFlow, QEMU, Linux Internals

Tools & Analytics: Git, GitHub, Google Analytics, beehiiv, Godot Engine (2D/3D)

Experience

codeabode

April 2024 – Present

Chief Executive Officer & Lead Developer

Plainsboro, NJ

- Architected a Python-based AI pipeline to dynamically generate and optimize personalized programming curriculums based on real-time student performance metrics
- Engineering an interactive 3D educational game in Godot to visually simulate core programming paradigms
- Lead product strategy and a team of 7+ to deliver 70+ hours of instruction to 100+ students, earning \$2,000 in revenue

AIOS – Rutgers University

May 2024 – Dec. 2025

Research Assistant

New Brunswick, NJ

- Co-authored AutoFlow, an automatic agentic workflow generator optimizing LLM task execution and multi-agent coordination
- Maintained open-source repository documentation for framework installation, system usage, and contributor guidelines
- Served as primary speaker at the AIOS Hackathon, presenting agentic infrastructure to an audience of 50+ developers

aquarc

June 2024 – July 2025

Chief Executive Officer & Lead Architect

Remote

- Built and launched an enterprise SAT prep platform scaling to 8,000+ questions using Go, React, and PostgreSQL
- Managed a distributed remote team of 10+ developers from product conception through agile deployment cycles
- Utilized Google Analytics and beehiiv to drive growth to 1,000+ active users and launch an internship newsletter

Projects

SwimIQ | *React, TensorFlow.js, Express, Groq Llama 4 Scout*

July 2026

- Developed a real-time computer vision web application using TensorFlow.js and MoveNet Thunder to perform in-browser pose estimation, tracking 17 body keypoints at 30 FPS
- Designed a hybrid edge/cloud AI architecture integrating Groq's Llama 4 Scout vision model to analyze uploaded videos and generate context-aware, terminology-specific coaching feedback
- Engineered a React frontend with live skeleton overlays, voice feedback, and a custom kinematic engine to identify stroke types, phase symmetry, and body rotation in highly variable real-world conditions
- Deployed at swim-iq.vercel.app

MeetTimeTracker | *Python, Gemini API, OCR, JSON*

Jan. 2026

- Engineered an AI agent utilizing the Gemini API for multimodal Optical Character Recognition (OCR) to automatically extract and parse unstructured physical swim meet results into a centralized database
- Designed custom prompt algorithms to resolve complex data row alignment errors and overlapping text structures within scanned images of medley relay results
- Automated the conversion of raw image data into highly accurate, structured JSON payloads, drastically reducing manual data entry for athletic tracking platforms

omOS | *C, x86 Assembly, QEMU*

Feb. 2024

- Developed a monolithic 16/32-bit operating system kernel from scratch featuring custom low-level process management
- Wrote a standalone C standard library and a VGA graphics driver integrated with a hardware interrupt keyboard handler
- Utilized QEMU for system emulation, kernel debugging, and low-level memory mapping

Leadership & Awards

New Jersey Privacy Movement | *Founder*

Jan. 2024 – Mar. 2024

- Authored a public petition for user data rights and collaborated with NJ State Senator Turner to draft a formal privacy bill

AIxEducation | *National Finalist & Co-Author*

Sep. 2024 – Nov. 2024

- Won 2nd Place Nationally for AI research simulating collaborative Montessori schools to replace traditional models
- Co-authored the Comprehensive Mental Health Support Act, securing co-sponsorship from State Senator Burgess

FBLA | *7th Place Regional Champion*

Sep. 2022 – June 2026

- Ranked 7th in Computer Problem Solving for Central NJ; raised \$150+ and collected 100+ shoes for charity drives

Community Safety Operations | *Certified Lifeguard*

May 2023 – Present

- Enforce pool safety protocols, manage risk prevention, and perform emergency water rescues and first-aid